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ADAPTIVE FINITE VOLUME METHODS FOR STEADY CONVECTION-DIFFUSION EQUATIONS WITH MESH OPTIMIZATION

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Abstract. An adaptive algorithm for steady convection-diffusion problems that combines a posteriori error estimation with conforming centroidal Voronoi Delaunay triangulations (CfCVDTs) is proposed and tested in two dimensional domains. Different from most current adaptive methods, this algorithm realizes mesh refinement and coarsening implicitly at each level by nodes insertion and redistribution. Especially, the nodes redistribution is implemented through the generation of CfCVDTs with some density function derived from the a posteriori error estimators for the problem. Numerical experiments show that the convergence rates achieved are almost the best obtainable using the linear finite volume discretizations and the resulting meshes always maintain high quality.

1. **Introduction.** In the past thirty years, adaptive methods have been playing more and more important roles in the solution process of partial differential equation (PDE) based problems in science and engineering. This methodology allows one to refine the mesh in the critical regions while remaining reasonably coarse in the rest of the domain. The essential components in most adaptive algorithms are the local error estimator and the mesh adaptivity technique. These two ingredients have been extensively studied beginning in the late 70's [3, 4] and followed by a vast literature, see [5, 8, 25, 26, 24, 28, 6] and references therein. Here we especially refer to [17, 18, 30, 31, 22, 27, 2, 7] for references on *a posteriori* error estimations and mesh refinements for steady convection-diffusion problems.

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The performance of adaptive methods for PDEs depends not only on the error estimators, but also on the techniques used for adaptively refining and generating meshes. In many if not most adaptive methods for PDEs, the meshes are only refined locally whenever some criterion based on a local error estimator is not satisfied. However, as pointed out in [6], a coarsening step is often necessary for the adaptive algorithms in order to obtain optimal convergence rates. There already exist many approaches to address this problem, for example, Laplacian smoothing [13] and moving mesh method [24, 32] are two of the popular ways. Recently, the mesh adaptivity approach based on the centroidal Voronoi tessellation (CVT) methodology was proposed in [20, 21] for numerical solution of the linear diffusion problems and the incompressible Navier-Stokes equations with finite element discretizations. This type of adaptive algorithm can distribute the nodes in some optimal way according to a posterior error estimates, so that the error of the resulting approximate solution is distributed equally over the elements and the resulting meshes always maintain very high quality. In this paper, we propose a similar mesh adaptive scheme for finite volume methods (central and upwind schemes) for solving the convection-diffusion problems (1), especially the convection-dominated cases. Our algorithm realizes mesh refinement and coarsening implicitly at each level by nodes insertion and redistribution. Especially, the nodes redistribution is implemented in a way to minimize the errors and improve the mesh quality.

The plan of the rest of the paper is as follows. The model convection-diffusion problem is first described in Section 1.1. In Section 2, we review and derive some specific a posteriori error estimators for the finite volume discretizations of the model equation. A remeshing scheme used to define our mesh adaptivity scheme will be discussed in Section 3. This meshing algorithm called *conforming centroidal Voronoi Delaunay triangulations* requires the use of a density function for deciding how grid nodes should be redistributed. We then show how that density function can be connected to the a posteriori error estimators in order to obtain certain optimality in error distribution and provide a description of our mesh adaptation algorithm. In Section 4, we perform several computational experiments to demonstrate the effectiveness and efficiency of our mesh adaptivity approach. Finally, some conclusions are drawn in Section 5.

1.1. **The model convection-diffusion problem.**

Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with a Lipschitz boundary $\partial\Omega$. Consider the following model convection-diffusion problem:

$$\left\{ \begin{array}{rcll} \nabla \cdot (-a\nabla u + \vec{\mathbf{b}}u) + ru & = & f, & \text{in } \Omega, \\ u & = & 0, & \text{on } \Gamma_D, \\ (-a\nabla u + \vec{\mathbf{b}}u)\cdot \vec{\mathbf{n}} & = & g, & \text{on } \Gamma_N^{\text{in}}, \\ -a\nabla u \cdot \vec{\mathbf{n}} & = & 0, & \text{on } \Gamma_N^{\text{out}}, \end{array} \right. \tag{1}$$

where $a(\mathbf{x}) \in H^1(\Omega)$ with $a(\mathbf{x}) \geq \tilde{a} > 0$, $\nabla \cdot \vec{\mathbf{b}}(\mathbf{x}) \in L^\infty(\Omega)$ represents the transport velocity, $\vec{\mathbf{n}}$ is the unit outer normal vector to $\partial\Omega$, $r(\mathbf{x}) \in L^\infty(\Omega)$ is a given reaction rate function and $f(\mathbf{x})$ denotes a given source function. The boundary $\partial\Omega$ of the domain is the union of the three parts: the Dirichlet boundary Γ_D, the inflow boundary $\Gamma_N^{\text{in}} = \{\mathbf{x} \in \partial\Omega : \vec{\mathbf{b}}(\mathbf{x}) \cdot \vec{\mathbf{n}}(\mathbf{x}) < 0\}$, and the outflow boundary $\Gamma_N^{\text{out}} = \{\mathbf{x} \in \partial\Omega : \vec{\mathbf{b}}(\mathbf{x}) \cdot \vec{\mathbf{n}}(\mathbf{x}) > 0\}$. Let $\Gamma_N = \Gamma_N^{\text{in}} \cup \Gamma_N^{\text{out}}$ and assume that Γ_D has positive measure.

In the following, we first review the general finite volume methods for the model problem (1), and then derive some results about explicit a posteriori error estimators used for our adaptive mesh refinement since they can be computed directly from the approximate solution.

2. Finite volume discretizations and error estimators.

2.1. Weak formulation of the model problem. Let V be the Hilbert space $H^1_D(\Omega) = \{v \in H^1(\Omega) \mid v = 0 \text{ on } \Gamma_D\}$. Define the bilinear form $A(\cdot,\cdot)$ on $V \times V$ as

$$A(u,v) = (a\nabla u - \vec{\mathbf{b}}u, \nabla v) + (ru, v) + \int_{\Gamma_N^{\text{out}}} (\vec{\mathbf{b}} \cdot \vec{\mathbf{n}}) uv \, ds, \tag{2}$$

where $(\cdot,\cdot)$ denotes the suppressed form of the inner product $(\cdot,\cdot)_{L^2(\Omega)}$. Then we can rewrite the problem (1) in the following weak form: find $u \in V$ such that

$$A(u,v) = F(v), \qquad \forall\, v \in V, \tag{3}$$

where $F(v) := (f, v) - \int_{\Gamma_N^{\text{in}}} gv \, ds$.

We also assume that the coefficients of the problem (1) satisfy the following V-elliptic (coercivity) and V-bounded (continuity) properties:

$$A(u,u) \ge c_0 \|u\|^2_{H^1}, \qquad \forall\, u \in V, \tag{4}$$

$$A(u,v) \le c_1 \|u\|_{H^1} \|v\|_{H^1}, \qquad \forall\, u,\ v \in V, \tag{5}$$

where c_0 and c_1 are two positive constants which are independent of u, $v \in V$.

The above two conditions (4) and (5) guarantee that the expression $[A(u,u)]^{1/2}$ is equivalent to the norm $\|\cdot\|_{H^1(\Omega)}$ in V, and that the variational problem (3) has a unique solution u in V. We also let $\|\cdot\|_E = (A(\cdot,\cdot))^{1/2}$ be the energy norm in V.

2.2. Central and upwind finite volume schemes. Assume that Ω is a polygonal domain with the boundary $\partial\Omega = \Gamma$ and $\mathcal{T}$ is a conforming triangulation (triangles) of Ω. Denote by h_T the diameter of the triangle $T \in \mathcal{T}$ (the length of the largest side of T) and by r_T the diameter of the largest circle that can be inscribed in T. Define the regularity ratio of the triangle T by $\kappa_T = h_T/r_T$. If there is a constant $\kappa > 0$ such that $\kappa_T \le \kappa$ for all $T \in \mathcal{T}$, then we say that the triangulation $\mathcal{T}$ of Ω is regular. It is worth noting that the assumption of the regularity permits partitions of the domain Ω into meshes that may contain elements of quite different sizes. This observation is very important for adaptive refinement. In the following, we will assume that $\mathcal{T}$ is regular, and that the mesh is aligned with the interfaces between Γ_D, Γ_D^{in}, and Γ_D^{out}.

Let $\mathcal{V} = \{\mathbf{x}_i \mid \mathbf{x}_i \text{ is a vertex of an element } T \in \mathcal{T}\}$ and $\mathcal{V}^0 = \{\mathbf{x}_i \in \mathcal{V} \mid \mathbf{x}_i \text{ is not on } \Gamma_D\}$. For a given vertex $\mathbf{x}_i$, let $\mathcal{N}_i = \{\mathbf{x}_j \mid \mathbf{x}_j \in T_j, \mathbf{x}_i \in T_j, \mathbf{x}_j \ne \mathbf{x}_i, T_j \in \mathcal{T}\}$ be the set of its neighbor vertices in $\mathcal{V}^0$ and N_i be the number of elements of $\mathcal{N}_i$. Let $N + M$ be the number of the elements in $\mathcal{V}$ where M is the number of vertices lying on Γ_D. Arrange order of the vertices so that $\{1, \cdots, N\}$ is for the those that do not lie on Γ_D and $\{N+1, \cdots, N+M\}$ for the those that do lie on Γ_D.

Based on the triangulation $\mathcal{T}$, we now construct a dual partition of Ω. For each vertex $\mathbf{x}_i$, let $\mathbf{x}_{i,j}$ $(j = 1, 2, \cdots, N_i)$ be the neighbor vertices of $\mathbf{x}_i$, $x_{i,j}$ with $j = 1, \cdots, N_i$ are ordered counterclockwise or clockwise, M_j be the middle point of the two vertices $\mathbf{x}_i$ and $\mathbf{x}_{i,j}$, and Q_j be a point inside the triangle $\Delta\mathbf{x}_i\mathbf{x}_{i,j}\mathbf{x}_{i,j+1}$. If $\mathbf{x}_i$ is an interior vertex, then we successively connect M_1, Q_1, M_2, Q_2, $\cdots$, M_{N_i}, Q_{N_i}, and M_1 to form a polygonal region D_i called the dual element of the vertex $\mathbf{x}_i$

(or the *control volume* of $\mathbf{x}_i$); otherwise, if $\mathbf{x}_i$ is on the boundary $\partial\Omega$, we successively connect $\mathbf{x}_i$, M_1, Q_1, M_2, Q_2, $\cdots$, M_{N_i}, Q_{N_i}, and $\mathbf{x}_i$ to form the dual polygonal region D_i. The set of dual polygonal elements $\mathcal{T}^* = \{D_i \mid \mathbf{x}_i \in \mathcal{V}\}$ is called the dual partition of Ω with respect to the triangulation $\mathcal{T}$. For adjacent vertices $\mathbf{x}_i$ and $\mathbf{x}_j$, we denote by $\Gamma_{ij} = \partial D_i \cap \partial D_j$ the common edge of the two dual elements D_i and D_j with the property $\partial D_i \backslash \Gamma = \sum_{j=1}^{N_i} \Gamma_{ij}$, by γ_{ij} the length of Γ_{ij}, and by $\vec{\nu}_{ij}$ the unit outer normal vector of Γ_{ij} from the dual element D_i.

For the construction of the dual element D_i, there are two common choices for the position of Q_j [23]. One is to choose Q_j to be the barycenter of the triangle $\Delta\mathbf{x}_i\mathbf{x}_{i,j}\mathbf{x}_{i,j+1}$, the other is to let Q_j be the circumcenter of $\Delta\mathbf{x}_i\mathbf{x}_{i,j}\mathbf{x}_{i,j+1}$. For the latter setting (circumcenter based), it is often required that all triangles of $\mathcal{T}$ are acute ones and in that case the dual partition are also called the Voronoi tessellation of Ω. It is worth noting that the analysis in this paper is valid for both choices of the control volume and so does our adaptive method although the first choice is used in our numerical experiments.

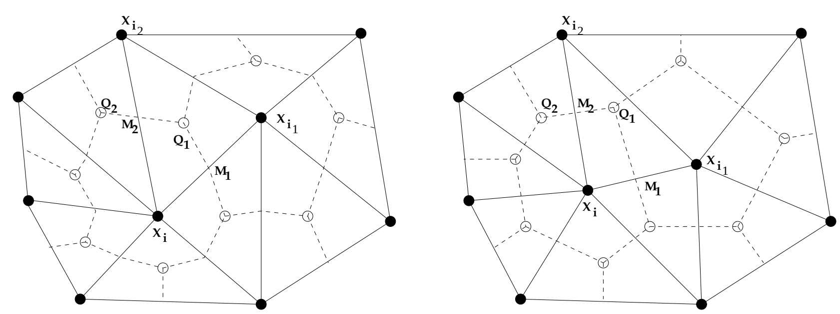

FIGURE 1. A triangulation and its dual partition (control volumes). Left: barycenter-based meshes where Q_i is the barycenter of the triangle $\mathbf{x}_i\mathbf{x}_{i,j}\mathbf{x}_{i,j+1}$; right: circumcenter-based meshes where Q_i is the circumcenter of the triangle $\mathbf{x}_i\mathbf{x}_{i,j}\mathbf{x}_{i,j+1}$ (Q_jQ_{j+1} is perpendicular to $\mathbf{x}_i\mathbf{x}_{i,j+1}$).

Let the patch $\widetilde{T} \in \Omega$ be the union of the triangle T and the other triangles in $\mathcal{T}$ that share at least one common vertex with T. We define $h_{\widetilde{T}} = \max_{T'\subset\widetilde{T}} h_{T'}$ and $r_{\widetilde{T}} = \min_{T'\subset\widetilde{T}} r_{T'}$, then the regularity of the patch $\widetilde{T}$ is measured by $\kappa_{\widetilde{T}} = h_{\widetilde{T}}/r_{\widetilde{T}}$. It is easy to see that the regularity of the triangulation $\mathcal{T}$ is automatically inherited by $\widetilde{\mathcal{T}} = \{\widetilde{T}\}$. The trial function space associated with $\mathcal{T}$ is defined by

$$V_h = \{v \in C(\overline{\Omega}) \mid v|_T \in \mathbb{P}_p(T), \quad \forall\, T \in \mathcal{T} \text{ and } v|_{\Gamma_D} = 0\},$$

where $\mathbb{P}_p(T)$ is the function space of polynomials of degree not larger than p on T. In this paper, we only consider the case $p = 1$, i.e., V_h is the continuous piecewise linear function space with respect to $\mathcal{T}$. Let $\{\phi_i(\mathbf{x}),\ i = 1, 2, \cdots, N\}$ be a basis of V_h such that $\phi_i(\mathbf{x})$ is equal to 1 at the vertex $\mathbf{x}_i$ and 0 at other vertices. Then for each $u_h \in V_h$, we have that $u_h(\mathbf{x}) = \sum\limits_{i=1}^{N} u_h(\mathbf{x}_i)\phi_i(\mathbf{x})$.

Now, we define the test function space V_h^* for the finite volume method by

$$V_h^* = \{v \in L^2(\Omega) \mid v|_{D_i} \equiv \textit{constant},\ i = 1, \cdots, N \ \text{ and } \ v|_{D_i} = 0,\ i = N{+}1, \cdots, M\}. \tag{6}$$

Let $\psi_i(\mathbf{x})$ be the characteristic function associated with D_i. Then $\{\psi_i(\mathbf{x}),\ i = 1, 2\cdots, N\}$ forms a basis of V_h^*. Denote by I_h and I_h^* the interpolation projectors from $C(\Omega)$ onto V_h and V_h^* respectively, such that for any $v \in C(\Omega)$, we have that

$$(I_h v)(\mathbf{x}) = \sum_{i=1}^{N} v(\mathbf{x}_i)\phi_i(\mathbf{x}), \quad (I_h^* v)(\mathbf{x}) = \sum_{i=1}^{N} v(\mathbf{x}_i)\psi_i(\mathbf{x}).$$

It is clear that $(I_h^*\phi_i)(\mathbf{x}) = \psi_i(\mathbf{x})$ and $V_h^* = \text{span}\{I_h^*\phi \mid \phi \in V_h\}$. Multiply the equation (1) by a $v_h \in V_h^*$ in both sides, integrate over Ω, and apply the Green's formula, we obtain

$$B(u, v_h) = L(v_h), \tag{7}$$

where $B(u, v_h) = a(u, v_h) + b(u, v_h) + c(u, v_h)$ with

$$\begin{aligned}
a(u, v_h) &= -\sum_{i=1}^{N} v_h(\mathbf{x}_i) \int_{\partial D_i \setminus \Gamma_N} (a\nabla u) \cdot \vec{\nu}\, ds, \\
b(u, v_h) &= \sum_{i=1}^{N} v_h(\mathbf{x}_i) \int_{\partial D_i \setminus \Gamma_N^{in}} (\vec{\mathbf{b}} \cdot \vec{\nu}) u\, ds, \\
c(u, v_h) &= (ru, v_h), \\
L(v_h) &= \sum_{i=1}^{N} v_h(\mathbf{x}_i) \left(\int_{D_i} f d\mathbf{x} - \int_{\partial D_i \cap \Gamma_N^{\text{in}}} g\, ds \right),
\end{aligned}$$

where $\vec{\nu}$ is the unit outer normal along ∂D_i.

The central finite volume discretization of the problem (1) is then given by the following variational problem: find $u_h \in V_h$ such that

$$B(u_h, v_h) = L(v_h), \qquad \forall\, v_h \in V_h^*. \tag{8}$$

The discretization scheme (8) is often used to solve non-convection-dominated problems. For the more interesting convection-dominated problems, the approximate solution u_h found by (8) often has nonphysical oscillations. To avoid this drawback, some special techniques are needed to approximate the convection term $b(u_h, v_h)$. Upwind scheme is one of the very popular treatments that is conservative and stable for this type of problems. Define

$$\beta_{ij} = \int_{\Gamma_{ij}} \vec{\mathbf{b}} \cdot \vec{\nu}_{ij}\, ds,$$

where ν_{ij} denotes the outer unit normal of the control volume D_i on Γ_{ij}. Let $\beta_{ij}^+ = \max(\beta_{ij}, 0)$ and $\beta_{ij}^- = \min(\beta_{ij}, 0)$, then we define the upwind approximation of $b(u_h, v_h)$ by

$$b_{up}(u_h, v_h) = \sum_{i=1}^{N} v_h(\mathbf{x}_i) \Big\{ \sum_{\Gamma_{ij} \setminus \Gamma_N} (\beta_{ij}^+ u_h(\mathbf{x}_i) + \beta_{ij}^- u_h(\mathbf{x}_j)) + \sum_{\Gamma_{ij} \cap \Gamma_N^{\text{out}}} \beta_{ij} u_h(\mathbf{x}_i) \Big\}. \tag{9}$$

Based on (8) and (9), an upwind finite volume discretization for the problem (1) then can be constructed as follows: find $u_h \in V_h$ such that

$$B_{up}(u_h, v_h) = L(v_h), \qquad \forall\, v_h \in V_h^*, \tag{10}$$

where $B_{up}(u_h, v_h) = a(u_h, v_h) + b_{up}(u_h, v_h) + c(u_h, v_h)$.

We denote by h the piecewise linear function with respect to $\mathcal{T}$ satisfying

$$h(\mathbf{x}_i) = \max_{T \in \mathcal{T} \text{ and } \mathbf{x}_i \in \overline{T}} h_T$$

for each vertex $\mathbf{x}_i \in \mathcal{T}$. We then have the following classic results [23]:

Theorem 2.1. *Assume that the coefficients of the problem (1) satisfy the V-elliptic (4) and the V-bounded (5) conditions. Assume that $u \in H^2(\Omega)$ be the analytic solution of the problem (3), and $u_h \in V_h$ be the solution of the equation (8) (central scheme) or (10) (upwind scheme). Then it holds*

$$\|u - u_h\|_E \leq C\|hu\|_{H^2(\Omega)}. \tag{11}$$

for some generic constant $C > 0$ independent of u and h.

2.3. **Explicit a posteriori error estimator.** In the following, we will briefly derive an explicit a posteriori error estimator for both central and upwind schemes, which is similar to the one presented in [7]. A major difference is that the upwind discretization of the convection term in our paper is different from the one used in [7]. The upwind scheme used here (9) is very general and more convenient for numerical implementation; and the explicit a posteriori error estimator can be derived together with the central scheme in a more unified and efficient way.

Let $e = u - u_h$ be the error of the approximate solution u_h of the upwind scheme (10). Integrating by parts, one can easily deduce

$$A(e,e) = (f - ru_h, e) + (\sigma_h, \nabla e) - \int_{\Gamma_N^{\text{out}}} (\vec{\mathbf{b}} \cdot \vec{\mathbf{n}})\ u_h\ e\ ds - \int_{\Gamma_N^{\text{in}}} ge\ ds, \tag{12}$$

$$(\sigma_h, \nabla e)_{L^2(T)} = \int_{\partial T} (\sigma_h \cdot \vec{\nu}_T) e\ ds - \int_T (\nabla \cdot \sigma_h) e\ d\mathbf{x}, \tag{13}$$

where $\sigma_h = -(a\nabla u_h - \vec{\mathbf{b}}u_h)$, and $\vec{\nu}_T$ is the outer unit normal vector to ∂T of the triangle T . Summing the equation (13) over all elements of $\mathcal{T}$ and inserting the derived equation into the equation (12) give us

$$\begin{aligned} A(e,e) &= (f - \nabla \cdot \sigma_h - ru_h, e) \\ &\quad + \sum_{\gamma \in \mathcal{E}} \int_\gamma [\sigma_h \cdot \vec{\nu}]_\gamma e ds - \int_{\Gamma_N^{\text{out}}} (a\nabla u_h \cdot \vec{\nu}) e\ ds - \int_{\Gamma_N^{\text{in}}} (g - \sigma_h \cdot \vec{\nu})\ e\ ds. \end{aligned} \tag{14}$$

Here $\mathcal{E} = \{\partial T \backslash \Gamma \mid T \in \mathcal{T}\}$, and $[\sigma_h \cdot \vec{\nu}]_\gamma$ denotes the jump of σ_h in the normal flux across the edge γ if T and T' share the common edge $\gamma \in \mathcal{E}$, i.e., $[\sigma_h \cdot \vec{\nu}]_\gamma = \sigma_h|_T \cdot \vec{\nu}_T + \sigma_h|_{T'} \cdot \vec{\nu}_{T'}$.

Let u_h^* be the function defined on Γ_{ij} such that

$$u_h^*|_{\Gamma_{ij} \cap \Gamma_N^{\text{out}}} = u_h(\mathbf{x}_i), \qquad u_h^*|_{\Gamma_{ij} \backslash \Gamma_N} = H(\beta_{ij}) u_h(\mathbf{x}_i) + (1 - H(\beta_{ij})) u_h(\mathbf{x}_j)$$

with the parameters β_{ij} given in (9) and $H(\mathbf{x})$ denoting the classical Heaviside function. Then the solution u_h of the upwind scheme (10) satisfies

$$\begin{aligned} \int_{\partial D_i \backslash \Gamma_N} (-a\nabla u_h + \vec{\mathbf{b}} u_h^*) \cdot \vec{\nu}\ ds &= \int_{D_i} (f - ru_h)\ d\mathbf{x} \\ &\quad - \int_{\partial D_i \cap \Gamma_N^{\text{in}}} (\vec{\mathbf{b}} \cdot \vec{\nu}) g\ ds - \int_{\partial D_i \cap \Gamma_N^{\text{out}}} (\vec{\mathbf{b}} \cdot \vec{\nu}) u_h^*\ ds. \end{aligned} \tag{15}$$

Note that (15) can be rewritten in the following equivalent form

$$\begin{aligned}\int_{\partial D_i\backslash\Gamma_N}\sigma_h\cdot\vec{\nu}ds &= \int_{D_i}(f-ru_h)d\mathbf{x}-\int_{\partial D_i\cap\Gamma_N^{\text{in}}}(\vec{\mathbf{b}}\cdot\vec{\nu})gds\\ &\quad-\int_{\partial D_i\cap\Gamma_N^{\text{out}}}(\vec{\mathbf{b}}\cdot\vec{\nu})u_h^*ds+\int_{\partial D_i\backslash\Gamma_N}(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)ds.\end{aligned}\tag{16}$$

Applying the Green's formula to the left hand side of the above equality over each subregion $T\cap D_i$ and summing the derived quantities, we obtain

$$\int_{\partial D_i\backslash\Gamma_N}\sigma_h\cdot\vec{\nu}\ ds = \int_{D_i}\nabla\cdot\sigma_h\ d\mathbf{x}-\sum_{\gamma\in\mathcal{E}_i}\int_\gamma[\sigma_h\cdot\vec{\nu}]_\gamma\ ds-\int_{\partial D_i\cap\Gamma_N}\sigma_h\cdot\vec{\nu}\ ds\tag{17}$$

where

$$\mathcal{E}_i=\{\partial(T\cap D_i)\backslash(\partial D_i\cup\Gamma_N)\ \mid\ T\in\mathcal{T}\}.$$

From (16) and (17), we deduce

$$\begin{aligned}&0\\ =\ &(f-\nabla\cdot\sigma_h-ru_h,\bar{e})+\sum_{\gamma\in\mathcal{E}}\int_\gamma[\sigma_h\cdot\vec{\nu}]_\gamma\bar{e}\ ds\\ &+\sum_{i=1}^N\bar{e}_i\int_{\partial D_i\backslash\Gamma_N}(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)ds\\ &-\int_{\Gamma_N^{\text{out}}}(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h^*-u_h)\bar{e}\ ds-\int_{\Gamma_N^{\text{out}}}(a\nabla u_h)\cdot\vec{\nu}\bar{e}\ ds\\ &-\int_{\Gamma_N^{\text{in}}}(\vec{\mathbf{b}}\cdot\vec{\nu})(g-\sigma_h\cdot\vec{\nu})\bar{e}\ ds\end{aligned}\tag{18}$$

where $\bar{e}=I_h^*e=\sum_{i=1}^N\bar{e}_i\psi_i(\mathbf{x})\in V_h^*$.

Subtracting the equation (18) from the equation (14), we obtain

$$\begin{aligned}A(e,e) &= (f-\nabla\cdot\sigma_h-ru_h,e-\bar{e})+\sum_{\gamma\in\mathcal{E}}\int_\gamma[\sigma_h\cdot\vec{\nu}]_\gamma(e-\bar{e})\ ds\\ &\quad-\int_{\Gamma_N^{\text{out}}}(a\nabla u_h\cdot\vec{\nu})(e-\bar{e})\ ds-\int_{\Gamma_N^{\text{in}}}(g-\sigma_h\cdot\vec{\nu})(e-\bar{e})\ ds\\ &\quad-\sum_{i=1}^N\bar{e}_i\int_{\partial D_i\backslash\Gamma_N}(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)\ ds-\int_{\Gamma_N^{\text{out}}}(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)\bar{e}\ ds\\ &= \sum_{i=1}^6 R_i.\end{aligned}\tag{19}$$

Now let us estimate the terms on the right hand side of the equation (19). In the following contexts, the constant C is a generic constant which does not depend on the partition $\mathcal{T}$, the exact solution u of the problem (1) and the solution u_h of the upwind scheme (10). The regularity of $\mathcal{T}$ allows us to use the trace theorem, the Poincaré and the Friedrich inequalities to get

$$\|e-\bar{e}\|_{L^2(D_i)} \leq Ch_i\|\nabla e\|_{L^2(D_i)},\tag{20}$$

$$\|e-\bar{e}\|_{L^2(\mathcal{E}_i)} \leq C(h_i^{-\frac{1}{2}}\|e-\bar{e}\|_{L^2(D_i)}+h_i^{\frac{1}{2}}\|\nabla(e-\bar{e})\|_{L^2(D_i)}),\tag{21}$$

where h_i denotes the diameter of D_i. By Cauchy-Schwarz inequality and the two inequalities (20) and (21), it is easy to obtain the estimates of $R_i(i=1,\cdots,6)$ as follows:

$$
\begin{aligned}
|R_1| &\leq C\sum_{i=1}^{N} h_i \|f-\nabla\cdot\sigma_h - ru_h\|_{L^2(D_i)}\|\nabla e\|_{L^2(D_i)}, && (22)\\
|R_2| &\leq C\sum_{i=1}^{N} h_i^{\frac{1}{2}} \|[\sigma_h\cdot\vec{\nu}]\|_{L^2(\mathcal{E}_i)}\|\nabla e\|_{L^2(D_i)}, && (23)\\
|R_3| &\leq C\sum_{\mathcal{E}_i\in\Gamma_N^{\text{out}}} h_i^{\frac{1}{2}} \|(a\nabla u_h)\vec{\nu}\|_{L(\mathcal{E}_i)}\|\nabla e\|_{L(D_i)}, && (24)\\
|R_4| &\leq C\sum_{\mathcal{E}_i\in\Gamma_N^{\text{in}}} h_i^{\frac{1}{2}} \|g-\sigma_h\cdot\vec{\nu}\|_{L(\mathcal{E}_i)}\|\nabla e\|_{L(D_i)}, && (25)\\
|R_6| &\leq C\sum_{\mathcal{E}_i\in\Gamma_N^{\text{out}}} h_i^{-\frac{1}{2}} \|(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)\|_{L^2(\mathcal{E}_i)}\|\nabla e\|_{L^2(D_i)}. && (26)
\end{aligned}
$$

The fact that the integral along Γ_{ij} occurs two times with the opposite directions $\vec{\nu}$ leads to

$$
\begin{aligned}
|R_5| &= \Big| -\frac{1}{2}\sum_{i=1}^{N}\sum_{j\in\mathcal{N}_i}\int_{\Gamma_{ij}}(\bar{e}-e)(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)\,ds\Big|\\
&\leq C\sum_{T\in\mathcal{T}} h_T^{\frac{1}{2}}\|(\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*)\|_{L^2(\mathcal{E}_T)}\|\nabla e\|_{L^2(T)},
\end{aligned}
\tag{27}
$$

where $\mathcal{E}_T = \cup_{\mathbf{x}_i\in T}((\partial(T\cap D_i)\cap\partial D_i)\backslash\Gamma)$. For any triangle $T\in\mathcal{T}$, let us define

$$
\eta_T = [(\eta_R)^2 + (\eta_E)^2 + (\eta_N)^2 + (\eta_E^{\text{up}})^2 + (\eta_E^{\text{out}})^2]^{1/2} \tag{28}
$$

with

$$
\begin{aligned}
\eta_R &:= \|h(f-\nabla\cdot\sigma_h - ru_h)\|_{L^2(T)},\\
\eta_E &:= \|h^{1/2}([\sigma_h\cdot\vec{\nu}])\|_{L^2(\partial T\cap\mathcal{E})},\\
\eta_N &:= \|h^{1/2}(a\nabla u_h\cdot\vec{\nu})\|_{L^2(\partial T\cap\Gamma_N^{\text{out}})} + \|h^{1/2}(g-\sigma_h\cdot\vec{\nu})\|_{L^2(\partial T\cap\Gamma_N^{\text{in}})},\\
\eta_E^{\text{up}} &:= \|h^{1/2}((\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*))\|_{L^2(\partial T)},\\
\eta_E^{\text{out}} &:= \|h^{-1/2}((\vec{\mathbf{b}}\cdot\vec{\nu})(u_h-u_h^*))\|_{L^2(\partial T\cap\Gamma_N^{\text{out}})}.
\end{aligned}
\tag{29}
$$

Based on the equality (19) and the estimates of R_i, $i=1,2,\cdots,6$, we can easily get the following result.

Theorem 2.2. *Assume that $u\in H^2(\Omega)$ be the exact solution of the problem (3), and $u_h\in V_h$ be the approximate solution of the upwind discretization scheme (10). Then there exists some generic constant $C>0$ such that*

$$
\|u-u_h\|_E \leq C\eta, \tag{30}
$$

where $\eta^2 = \sum_{T\in\mathcal{T}}\eta_T^2$.

Remark 1. The estimator η_E^{out} depends on the upwind value u_h^* on the outflow boundary Γ_N^{out}. The local error estimator η_E^{out} can also be changed to

$$
\eta_E^{\text{out}} = \|h^{1/2}((\vec{\mathbf{b}}\cdot\vec{\nu})(\nabla u_h\cdot\vec{\tau}))\|_{L^2(\Gamma_N^{\text{out}})}.
$$

where $\vec{\tau}$ is the unit tangent vector on Γ_N^{out}.

Remark 2. The same result as stated in Theorem 2.2 also holds for the approximate solution obtained from the central finite volume scheme (8). Especially, we have $\eta_E^{\text{up}} = \eta_E^{\text{out}} = 0$ in this case.

The inequality (30) shows that the true error e_h can be bounded from above in terms of the local error estimators η_R, η_E, η_N, η_E^{up}, and η_E^{out}, i.e., when they are small, the true error e must be small too. This property is referred to as the *reliability* of this error estimator. Adaptive numerical methods generally also need the fact that the true error e is also locally bounded from below by the local error estimators . This type of property is referred to as the *efficiency* of the error estimators. By using properly chosen bubble functions, the efficiency of these explicit the *a posteriori* error estimators can be proved for the central finite volume method (8). Details can be found in [7] and the references cited therein. For the upwind scheme, this is still an open problem.

3. Adaptive finite volume method.

The main idea of our adaptive finite volume method can be divided into two steps at each level: first, refine the current mesh according to the explicit local a posteriori error estimator (30), and then optimize the mesh (i.e., redistribute the nodes) using the so-called *conforming centroidal Voronoi Delaunay triangulation* (CfCVDT) [19, 20] with some density function determined from the error estimator.

3.1. Conforming centroidal Voronoi Delaunay triangulations.

Given an open convex domain $\Omega \in \mathbb{R}^d$ and a set of distinct points $\{\mathbf{x}_i\}_{i=1}^n \subset \Omega$, define the corresponding Voronoi region V_i, $i = 1, \dots, n$, for each point $\mathbf{x}_i$, $i = 1, \dots, n$, by

$$V_i = \big\{\mathbf{x} \in \Omega \ \mid\ |\mathbf{x} - \mathbf{x}_i| < |\mathbf{x} - \mathbf{x}_j| \quad \text{for } j = 1, \dots, n \quad \text{and} \quad j \neq i\big\}.$$

It is clear that $\mathbf{x}_i \in V_i$, $V_i \cap V_j = \emptyset$ for $i \neq j$, and $\cup_{i=1}^n \overline{V}_i = \overline{\Omega}$, so $\{V_i\}_{i=1}^n$ is a tessellation of Ω. We refer to $\{V_i\}_{i=1}^n$ as the *Voronoi tessellation* (VT) of Ω associated with the point set $\{\mathbf{x}_i\}_{i=1}^n$. The point $\mathbf{x}_i$ is called the *generator* of the *Voronoi region* $V_i \subset \Omega$. It is well known that the dual tessellation to a Voronoi tessellation of Ω is called a *Delaunay triangulation* (DT). Note that the vertices of the Voronoi regions V_i's consist of circumcenters of the corresponding Delaunay triangles. Given a density function $\rho(\mathbf{x})$ defined on Ω, for any region $V \subset \Omega$, we define the *mass centroid* $\mathbf{x}^*$ of V by

$$\mathbf{x}^* = \frac{\int_V \mathbf{y}\rho(\mathbf{y})\,\mathrm{d}\mathbf{y}}{\int_V \rho(\mathbf{y})\,\mathrm{d}\mathbf{y}}. \tag{31}$$

In general, the centers of mass of the Voronoi sets do not coincide with the generators of the Voronoi sets, i.e., $\mathbf{x}_i \neq \mathbf{x}_i^*$, $i = 1, \dots, n$, where $\mathbf{x}_i^*$ is the mass centroid of the Voronoi region V_i.

Definition 3.1. [9] A Voronoi tessellation $\{(\mathbf{x}_i, V_i)\}_{i=1}^n$ of Ω is called a *centroidal Voronoi tessellation (CVT)* with respect to $\rho(y)$ if and only if the points $\{\mathbf{x}_i\}_{i=1}^n$, which serve as the generators of the associated Voronoi tessellation $\{V_i\}_{i=1}^n$, are also the mass centroids of those regions, i.e.,

$$\mathbf{x}_i = \mathbf{x}_i^* \quad \text{for } i = 1, \dots, n.$$

The corresponding Delaunay triangulation is referred to as a *centroidal Voronoi-Delaunay triangulation (CVDT).*

Given any set of points $\{\widetilde{\mathbf{x}}_i\}_{i=1}^n$ on Ω and any tessellation $\{\widetilde{V}_i\}_{i=1}^n$ of Ω, we define the corresponding *energy* by

$$\mathbf{K}\big(\{(\widetilde{\mathbf{x}}_i, \widetilde{V}_i)\}_{i=1}^n\big) = \sum_{i=1}^n \int_{\widetilde{V}_i} \rho(\mathbf{y})|\mathbf{y} - \widetilde{\mathbf{x}}_i|^2\,\mathrm{d}\mathbf{y}\,. \tag{32}$$

A very useful property of CVTs is that the energy is equally distributed over the Voronoi regions V_is in an asymptotic way if $\{(\mathbf{x}_i, V_i)\}_{i=1}^n$ is a CVT. In the one-dimensional case, it was shown in [9] that $\mathbf{K}_{V_i} \approx \mathbf{K}/n$ for $i = 1, \ldots, n$, where $\mathbf{K}_{V_i} = \int_{V_i} \rho(\mathbf{x})\|\mathbf{x} - \mathbf{x}_i\|^2 \,\mathrm{d}\mathbf{x}$ and $\mathbf{K} = \sum_{i=1}^n \mathbf{K}_{V_i}$. For higher-dimensional cases, this property is only a conjecture [16] but its validity has been verified through extensive numerical studies and is widely assumed in practical applications such as vector quantization. Based on this energy equi-partition property, CVTs possess many important geometric features. For a constant density function, the generators $\{\mathbf{x}_i\}_{i=1}^n$ are uniformly distributed; the Voronoi regions $\{V_i\}_{i=1}^n$ are all almost of the same size, and in the two-dimensional case, most of them are (nearly) congruent convex hexagons [9]. For a non-constant density function, the generators $\{\mathbf{x}_i\}_{i=1}^n$ are still locally uniformly distributed and from the energy equi-partition property, it is easy to deduce that, asymptotically, for some constant C,

$$\mathbf{K}_{V_i} \approx C\rho(\mathbf{x}_i) h_{V_i}^{d+2} \qquad \text{and} \qquad \frac{h_{V_i}}{h_{V_j}} \approx \Big(\frac{\rho(\mathbf{x}_j)}{\rho(\mathbf{x}_i)}\Big)^{\frac{1}{d+2}}, \tag{33}$$

where h_{V_i} denotes the diameter of V_i and d is the dimension of Ω. Therefore in principle, one could control the distribution of generators to obtain an equal distribution of the error by connecting the density function $\rho(\mathbf{x})$ to an *a posteriori* error estimator.

An often used algorithm for constructing CVT/CVDT is the so-called the *Lloyd's algorithm* which is essentially a simple iteration between constructing Voronoi diagrams and mass centroids, see [9] for details. In order to use CVDT meshes within a discretization method for a PDE, some of the CVDT nodes should be constrained to lie on the boundary so that the boundary conditions of the PDE problem can be enforced [10]. A nice modification of CVDT was recently proposed in [19, 20] and is briefly reviewed in the following. Assume that Ω is compact and the domain boundary $\partial\Omega$ is piecewise smooth. The set of singular points of Ω, e.g., corners, is denoted by $P_S = \{\mathbf{z}_i\}_{i=1}^k$. Let

$$P_I = \{\mathbf{x}_i \mid \overline{V}_i \cap \partial\Omega = \emptyset\} \qquad \text{and} \qquad P_B = \{\mathbf{x}_i \mid \overline{V}_i \cap \partial\Omega \neq \emptyset\},$$

so that P_I, the set of *interior Voronoi generators*, denotes the set of generators whose Voronoi regions do not intersect the boundary $\partial\Omega$, and P_B, the set of *boundary Voronoi generators*, denotes the set of generators whose Voronoi regions do intersect the boundary $\partial\Omega$. Denote by $\mathbf{Proj}(\mathbf{x})$ the closest point on the boundary $\partial\Omega$ to $\mathbf{x}$.

Definition 3.2. [20] A Voronoi Tessellation $\{(\mathbf{x}_i, V_i)\}_{i=1}^n$ of Ω is called a *conforming centroidal Voronoi tessellation (CfCVT)* if and only if the following properties are satisfied:

- $P_S \subset \{\mathbf{x}_i\}_{i=1}^n$;
- $\mathbf{x}_i = \mathbf{x}_i^*$ for $\mathbf{x}_i \in P_I$;
- $\mathbf{x}_i = \mathbf{Proj}(\mathbf{x}_i^*)$ for $\mathbf{x}_i \in P_B \backslash P_S$.

The corresponding dual triangulation is then called a *conforming centroidal Voronoi Delaunay triangulation (CfCVDT)*.

An effective algorithm called the *Modified Lloyd's algorithm* for constructing such CfCVDTs in two dimensional space was implemented in the software package "MESHGEN" [19] based on the *constrained Delaunay triangulation* (CDT) process [29]. This algorithm takes a domain Ω, a triangular mesh $\mathcal{T}_0$ of Ω, and a density function ρ as input, and the output is a new mesh $\mathcal{T}$ of Ω which has the same number of vertices as that of the vertices of the mesh $\mathcal{T}_0$ and is also a CfCVDT. We will use the notation $\mathcal{T}$=CfCVDT($\mathcal{T}_0$,Ω,ρ) to represent this Algorithm. Fig. 2 presents some sample CfCVDT meshes with different domains and density functions. Details can be found in [19].

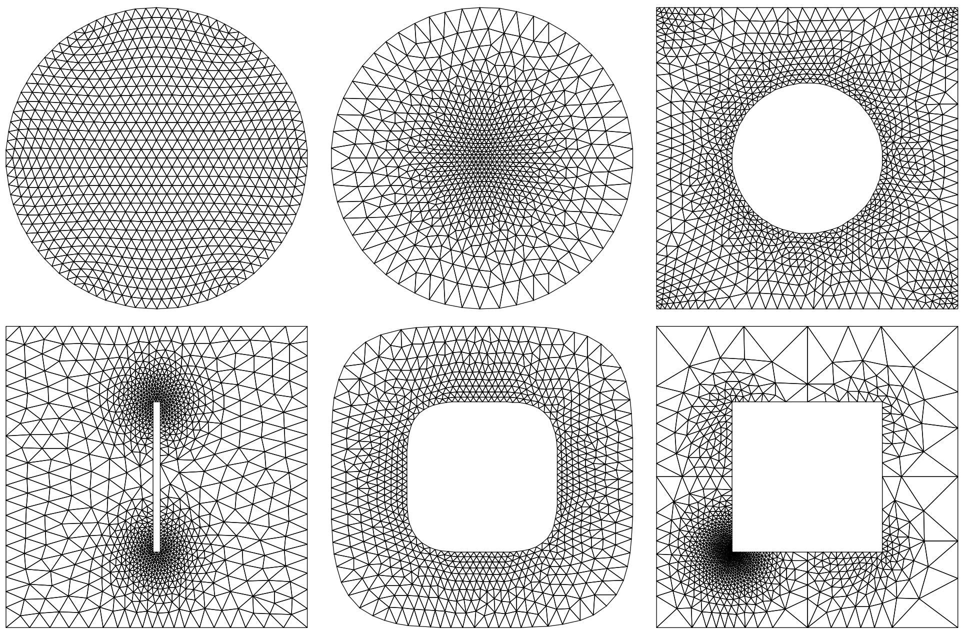

FIGURE 2. Some sample CfCVDT meshes with various domains and density functions.

3.2. **Adaptive meshing scheme.** Let us focus on two dimensional spaces. Let $\mathcal{T}^{(\ell)}$ denote the triangulation of Ω with the vertices $\{\mathbf{x}_i^{(\ell)}\}_{i=1}^{n^{(\ell)}}$ at the refinement level ℓ. Let $\eta_T^{(\ell)}$ ($T \in \mathcal{T}^{(\ell)}$) represent the corresponding local error estimators at level ℓ. In order to minimize $(\eta^{(\ell)})^2 = \sum_{T\in\mathcal{T}^{(\ell)}} (\eta_T^{(\ell)})^2$, we need to distribute $(\eta_T^{(\ell)})^2$ equally over all triangles of $\mathcal{T}^{(\ell)}$.

First set

$$\widetilde{\rho}_T^{(\ell)} = \frac{(\eta_T^{(\ell)})^2}{h_T^4}, \tag{34}$$

then define the piecewise linear function (with respect to $\mathcal{T}^{(\ell)}$) $\rho^{(\ell+1)}$ on Ω determined by

$$\rho^{(\ell+1)}(\mathbf{x}_i^{(\ell)}) = \frac{\sum_{T\in S_i} \widetilde{\rho}_T^{(\ell)}}{\mathrm{card}(S_i)} \tag{35}$$

for the vertex $\mathbf{x}_i^{(\ell)}$ of $\mathcal{T}^{(\ell)}$, where $S_i = \{T \in \mathcal{T}^{(\ell)} \mid \mathbf{x}_i^{(\ell)} \in \overline{T}\}$, and card$(S_i)$ is the element number in the set S_i. If the solution u of (1) is in $H^2(\Omega)$, then it is easy to find

$$(\eta_T^{(\ell)})^2 \approx C|u|^2_{H^2(\Omega)} h_T^4, \tag{36}$$

where $|u|_{H^2(\Omega)}$ is the semi-norm of u in $H^2(\Omega)$, and C is a constant which does not depend on T.

According to the conjectured properties of CVT/CVDT in (33), for 2-dimensional problem ($d = 2$), we have

$$\frac{h_{V_i}}{h_{V_j}} \approx \Big(\frac{\rho(\mathbf{x}_j)}{\rho(\mathbf{x}_i)}\Big)^{1/4}$$

for a CVT $\{(\mathbf{x}_i, V_i)\}_{i=1}^n$ of Ω with respect to the density function ρ. It is then not difficult to verify that the CfCVDT mesh $\mathcal{T}^{(\ell+1)}$ generated by the density function $\rho^{(\ell+1)}$ will approximately have the property

$$(\eta_{T_i}^{(\ell+1)})^2 \approx (\eta_{T_j}^{(\ell+1)})^2 \tag{37}$$

for triangles T_i and $T_j \in \mathcal{T}^{(\ell+1)}$, i.e., the principle of equidistribution of errors is approximately met. We also note that, for the general d-dimensional problems, the density function $\widetilde{\rho}_T^{(\ell)}$ should be set to

$$\widetilde{\rho}_T^{(\ell)} = \frac{(\eta_T^{(\ell)})^{(d+2)/2}}{h_T^{d+2}}, \tag{38}$$

see [20] for details.

Now let us present our CfCVDT-based adaptive meshing algorithm. Let $\{E_i\}_{i=1}^{k^{(\ell)}}$ denote the set of edges of the ℓ-th level triangulation $\mathcal{T}^{(\ell)}$. Set $\rho_{E_i} = \rho(\mathbf{x}_{E_i})$ for any density function ρ, where $\mathbf{x}_{E_i}$ denotes the midpoint of the edge E_i.

Algorithm 1. *Given a domain Ω, an integer $N_{max} > 0$ (the maximum allowable number of mesh vertices), an integer L_{max} (the maximum allowable levels of refinements), and a parameter $0 < \theta \le 1$, do the followings to obtain the refinement triangulations $\mathcal{T}^{(\ell)}$ $(l = 0, 1, \cdots, L_{max})$.*

0. *Generate an initial coarse triangulation $\mathcal{T}^{(0)}$ of Ω using CDT or some other means. Let $n^{(0)}$ denote the number of vertices of $\mathcal{T}^{(0)}$ and set $\ell = 0$.*
1. *Solve the convection-diffusion equation using the FVM on $\mathcal{T}^{(\ell)}$. If $\ell > L_{max}$ or $n^{(\ell)} > N_{max}$, terminate; otherwise, go to step 2.*
2. *Determine the local error estimator $\eta_T^{(\ell)}$ for all $T \in \mathcal{T}^{(\ell)}$.*
3. *Construct the density function $\rho = \rho^{(\ell+1)}$.*
4. *Determine $\{\rho_{E_i}\}_{i=1}^{k^{(\ell)}}$ and sort them in decreasing order.*
5. *Add $\{\mathbf{x}_{E_i}\}_{i=1}^{k_\theta}$ into the triangulation $\mathcal{T}^{(\ell)}$, where*

$$k_\theta = \max\Big\{k^* \;\Big|\; \sum_{i=1}^{k^*} \rho_{E_i} < \theta \sum_{i=1}^{k^{(l)}} \rho_{E_i}\Big\},$$

and then form, using CDT, the new intermediate triangulation $\widetilde{\mathcal{T}}^{(\ell+1)}$ with $n^{(\ell+1)} = n^{(\ell)} + k_\theta$ vertices.

6. *Optimize $\widetilde{\mathcal{T}}^{(\ell+1)}$ to obtain $\mathcal{T}^{(\ell+1)} = CfCVDT(\widetilde{\mathcal{T}}^{(\ell+1)}, \Omega, \rho)$.*
7. *Set $\ell \leftarrow \ell + 1$, then go to the step 1.*

The parameter θ in Algorithm 1 is used here to control the refinement process [8]. The most time-consuming step in the calculations of $\rho^{(\ell+1)}(\mathbf{x})$ for each $\mathbf{x} \in \Omega$ is the nearest-neighbor-search operation. This task can be achieved efficiently using the software package "ANN"[1].

Remark 3. This mesh adaptation strategy can be easily generalized and adapted to higher-order finite volume approximations, for example, the use of higher-order piecewise polynomial spaces and up-winding schemes. For the corresponding change on the density function (38), see [20] for some relevant discussions.

4. **Numerical experiments.** In this section, we will test several numerical examples to demonstrate the effectiveness and efficiency of our CfCVDT-based adaptive finite volume method. The problems we choose are typical convection-diffusion problems with some type of singularities. We set $L_{max} = 15$, N_{max} =20,000, and $\theta = 0.4$ for Algorithm 1. For our adaptive methods, the convergence rate CR with respect to the norm $\|\cdot\|$ at the refinement level ℓ is roughly computed by

$$\mathrm{CR} = \frac{2\log(\|e_{h,\ell}\|/\|e_{h,\ell-1}\|)}{\log(n_{\ell-1}/n_\ell)}, \tag{39}$$

where n_ℓ denotes the number of nodes and $e_{h,\ell}$ denotes the error $u - u_h$ at the refinement level ℓ. The quality measure $q(T)$ [14] is defined to be twice the ratio of the radius R_T of the largest inscribed circle and the radius r_T of the smallest circumscribed circle, i.e.,

$$q(T) = 2\frac{R_T}{r_T} = \frac{(b+c-a)(c+a-b)(a+b-c)}{abc} \tag{40}$$

where a, b and c are side lengths of the triangle T. For a given triangulation $\mathcal{T}$, we define

$$q_{min} = \min_{T\in\mathcal{T}} q(T) \qquad \text{and} \qquad q_{avg} = \frac{1}{\mathrm{card}(\mathcal{T})}\sum_{T\in\mathcal{T}} q(T). \tag{41}$$

Here, q_{min} measures the quality of the worst triangle and q_{avg} measures the average quality of the mesh $\mathcal{T}$. In order to evaluate the distribution of the local error estimator η_T over all elements of $\mathcal{T}$, we define the normalized standard deviation STD_η by

$$STD_\eta = \frac{\sqrt{\sum_{T\in\mathcal{T}}(\eta_T - \mathbf{E}\eta)^2/\mathrm{card}(\mathcal{T})}}{\mathbf{E}\eta}, \tag{42}$$

where $\mathbf{E}\eta$ denotes the expectation of η_T, i.e., $\mathbf{E}\eta = \sum_{T\in\mathcal{T}}\eta_T/\mathrm{card}(\mathcal{T})$. We also note that the barycenter-based control volume was used in our finite volume discretization.

4.1. **Problem with geometric singularity.** The first illustrative example is given as follows.

Example 1. *Let $\Omega = \Omega_{R_1}\cup\Omega_{R_2}$ where $\Omega_{R_1} = [-1,0]\times[-1,1]$ and $\Omega_{R_2} = [0,1]\times[0,1]$ so that Ω is a non-convex, Γ-shaped region which often induces a geometrically based singularity in the solution of the PDE at the origin $(0,0)$. We set $a(x,y) = 1$, $\vec{\mathbf{b}}(x,y) = (1,1)$ and $r(x) = 1$, then it is a regular convection-diffusion equation. We use the polar coordinates (r,θ) instead of the Cartesian coordinates $(x,y) = (r\cos\theta, r\sin\theta)$ to describe the exact solution u which is chosen to be*

$$u(r,\theta) = s\left(\frac{r-\delta_1}{\delta_2-\delta_1}\right) r^{2/3}\sin\left(\frac{2}{3}\theta\right) + w(r\cos\theta, r\sin\theta) \tag{43}$$

with $\delta_1 = 0.02$ and $\delta_2 = 0.25$, where s is the cut-off function

$$s(t) = \begin{cases} 1, & t<0,\\ -6t^5+15t^4-10t^3+1, & 0\le t\le 1,\\ 0, & t>1,\end{cases}$$

and $w(x,y) = (x-x^3)(y^2-y^4)$. Note that w is a smooth function with $w|_{\partial\Omega} = 0$. The homogeneous boundary condition $u = 0$ is used and the source function f is determined from u so that the target equation is satisfied.

Note that the exponent and angle factor $2/3$ in the exact solution (43) emulates the typical singular behavior of solutions of the target problem in the Γ-shaped domain that has an interior angle equal to $3\pi/2$ [15]. It is then easy to show that the exact solution $u \in H^{\frac{5}{3}-\epsilon}(\Omega)$ for any $\epsilon > 0$ and has a strong singularity at the origin. The central scheme was used for discretization to solve this problem.

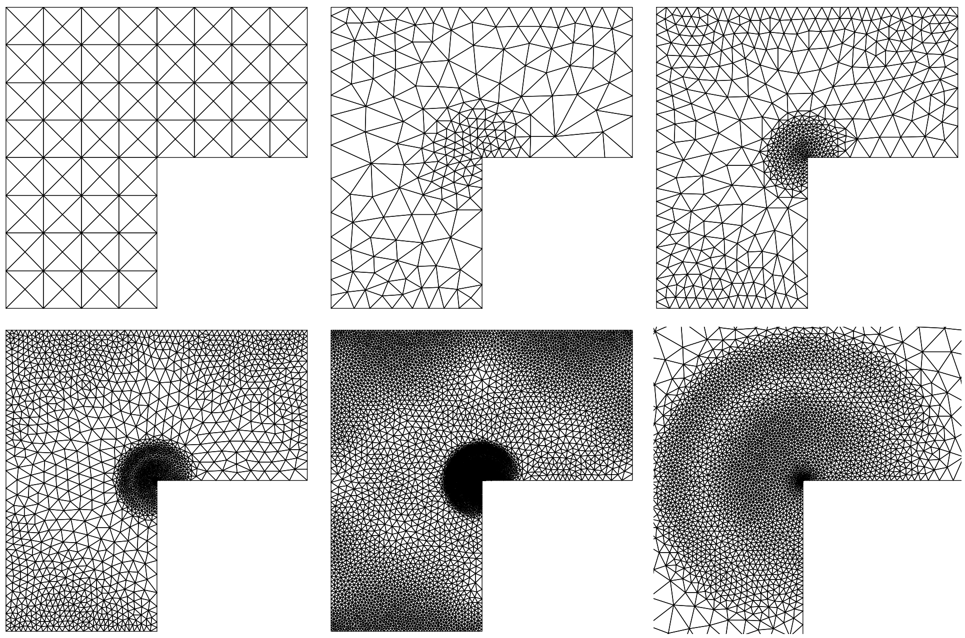

FIGURE 3. Refined meshes at some levels generated by the CfCVDT-based adaptive method for Example 1. Top-left: Initial mesh; top-middle, top-right, bottom-left and bottom-middle: adaptively refined meshes at levels 2, 4, 6, 8; bottom-right: a zoomed portion of level 8.

Level ℓ	n_ℓ	q_{min}	q_{avg}	h_{max}/h_{min}
0	113	0.828	0.828	1.00
1	174	0.725	0.939	3.18
2	217	0.591	0.915	6.55
3	351	0.575	0.932	12.59
4	621	0.513	0.926	17.61
5	1110	0.447	0.931	26.35
6	2026	0.490	0.939	31.80
7	3784	0.503	0.943	41.45
8	7160	0.536	0.943	86.92
9	13629	0.549	0.944	127.03
10	26011	0.572	0.944	293.09

TABLE 1. Mesh quality at different levels for Example 1.

The initial coarse mesh (the input for step 1 of Algorithm 1) used for Example 1 is a uniform Cartesian grid consisting of 113 nodes in the Γ-shaped domain, see Fig. 3.

Level ℓ	$\|e_{h,\ell}\|_{L^\infty(\Omega)}$	CR	$\|e_{h,\ell}\|_{L^2(\Omega)}$	CR	$\|e_{h,\ell}\|_{H^1(\Omega)}$	CR	STD_η
0	1.3169e-01	–	2.5802e-02	–	4.0522e-01	–	0.568
1	9.3896e-02	1.57	1.5708e-02	2.30	2.9048e-01	1.54	1.233
2	2.8724e-02	10.72	6.4370e-03	8.07	1.7438e-01	4.62	0.610
3	1.7072e-02	2.16	3.2727e-03	2.81	1.1821e-01	1.62	0.406
4	1.0386e-02	1.74	1.8252e-03	2.05	8.5768e-02	1.12	0.355
5	7.2812e-03	1.22	1.0079e-03	2.04	6.2018e-02	1.12	0.299
6	4.7168e-03	1.44	5.2255e-04	2.18	4.4654e-02	1.10	0.239
7	3.0272e-03	1.42	3.0070e-04	1.76	3.1601e-02	1.10	0.214
8	1.8840e-03	1.48	1.6324e-04	1.91	2.2930e-02	1.01	0.181
9	1.0838e-03	1.71	9.5608e-05	1.66	1.6425e-02	0.94	0.172
10	6.1109e-04	1.77	5.0473e-05	1.98	1.1853e-02	1.01	0.162

TABLE 2. Solution errors and convergence rates at different levels for Example 1.

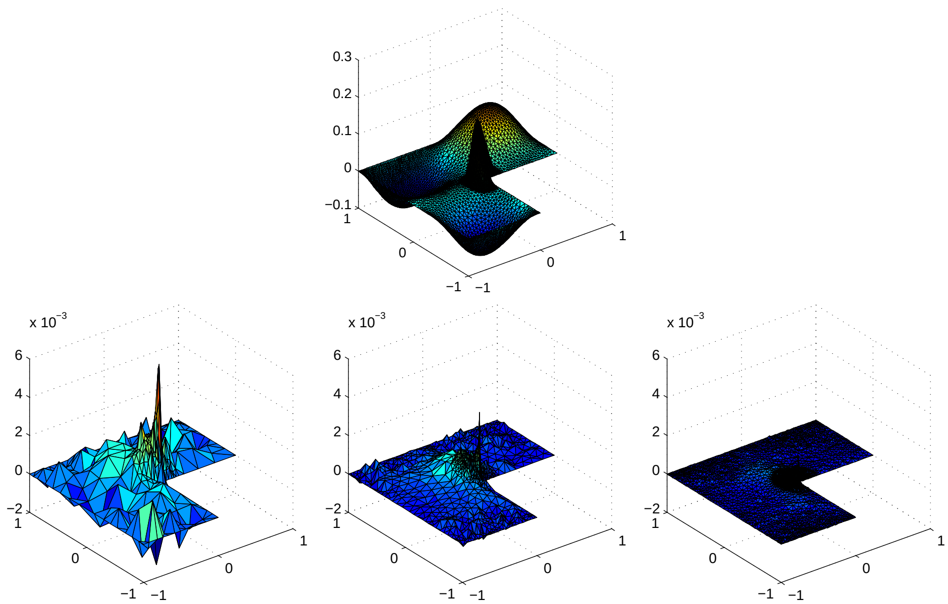

FIGURE 4. Plots of an approximate solution u_h and error distributions $e_{h,\ell}$ for Example 1. Top: plot of the approximate solution u_h at level 8; bottom (from left to right): error distributions $e_{h,\ell}$ at the refinement levels 4, 6 and 8.

This figure also shows adaptively refined CfCVDT meshes at some levels generated using Algorithm 1. It is clear that the distributions of nodes in the adaptive meshes show the accumulation of nodes at the origin where the exact solution has a strong singularity. The CfCVDT-based adaptive meshes are "optimal" partially in the sense that all triangles remain well-shaped at all refinement levels. The observation from the zoomed portion of level 8 in Fig. 3 is supported by the values of q_{min} and q_{avg} given in Table 1, even though the mesh size varies greatly over the Ω, e.g., h_{max}/h_{min} in Table 1 reaches 293.09 at the last level. Table 2 contains the information about solution errors, and convergence rates at different refinement levels for Example 1. It is well-known that if the uniform mesh refinement is used, the convergence rate will roughly be 4/3 measured in L^2 norm and 2/3 in H^1 norm. However, we get approximately the second order of convergence measured in L^2 norm and the first order in H^1 norm for our adaptive algorithm, the best obtainable. The converge rate in L^∞ norm also reaches more than 1.7 at the last two levels.

Another optimal property of CfCVDT-based adaptive methods being conjectured is the asymptotical equidistribution of the errors over Ω. To verify this, we plot, in Fig. 4, a representative approximate solution u_h and the errors $e_{h,\ell}$ at different refinement levels; it can also be verified by the values of the normalized standard deviation STD_η in Table 2 that are gradually decreasing along the refinement.

4.2. **Solution with a jump.** The second illustrative example is given as follows.

Example 2. *In this example, the domain is chosen to be the unit square* $\Omega = [0,1]\times[0,1]$. *We set the transport vector* $\mathbf{b} = (y, -x)$ *and the diffusion coefficient* $a = \epsilon$. *The boundary condition is given by*

$$\begin{cases} u = 0, & \text{on } \Gamma_D^1 = \{(x,y) \mid y = 1 \text{ and } 0 \le x \le 1\} \\ & \qquad \cup\{(x,y) \mid x = 0 \text{ and } 0.7 < y \le 1\}, \\ u = 1, & \text{on } \Gamma_D^2 = \{(x,y) \mid x = 0 \text{ and } 0 \le y \le 0.7\}, \\ \epsilon\dfrac{\partial u}{\partial \mathbf{n}} = 0, & \text{on } \Gamma_N = \{(x,y) \mid y = 0 \text{ and } 0 \le x \le 1\} \\ & \qquad \cup\{(x,y) \mid x = 1 \text{ and } 0 < y \le 1\}. \end{cases}$$

We also set $r = 0$ *and* $f = 0$, *then the analytic solution tends to*

$$u = \begin{cases} 1, & \sqrt{x^2+y^2} < 0.7, \\ 0, & \sqrt{x^2+y^2} > 0.7 \end{cases}$$

when $\epsilon \to +0$. *In the numerical experiments, we let* $\epsilon = 10^{-5}$.

It can be shown that the exact solution for this example only belongs to $L^\infty(\Omega)$ and has a jump from 0 to 1 around a quarter of the circle $\{(x,y) \mid x \ge 0, y \ge 0, x^2+y^2 = 0.7^2\}$ [17], where the strong singularity will cause heavy refinements.

Level ℓ	n_ℓ	q_{min}	q_{avg}	h_{max}/h_{min}
0	81	0.828	0.828	1.00
1	104	0.694	0.934	6.79
2	158	0.431	0.925	15.95
3	242	0.427	0.884	26.49
4	371	0.364	0.911	113.68
5	611	0.288	0.921	208.50
6	931	0.270	0.919	426.90
7	1443	0.251	0.929	435.23
8	2326	0.383	0.930	512.11
9	3773	0.291	0.935	971.63
10	6079	0.211	0.937	1262.69
11	9782	0.327	0.941	1571.22
12	16088	0.339	0.890	2112.83
13	27140	0.325	0.944	2447.40

TABLE 3. Mesh quality at different refinement levels for Example 2.

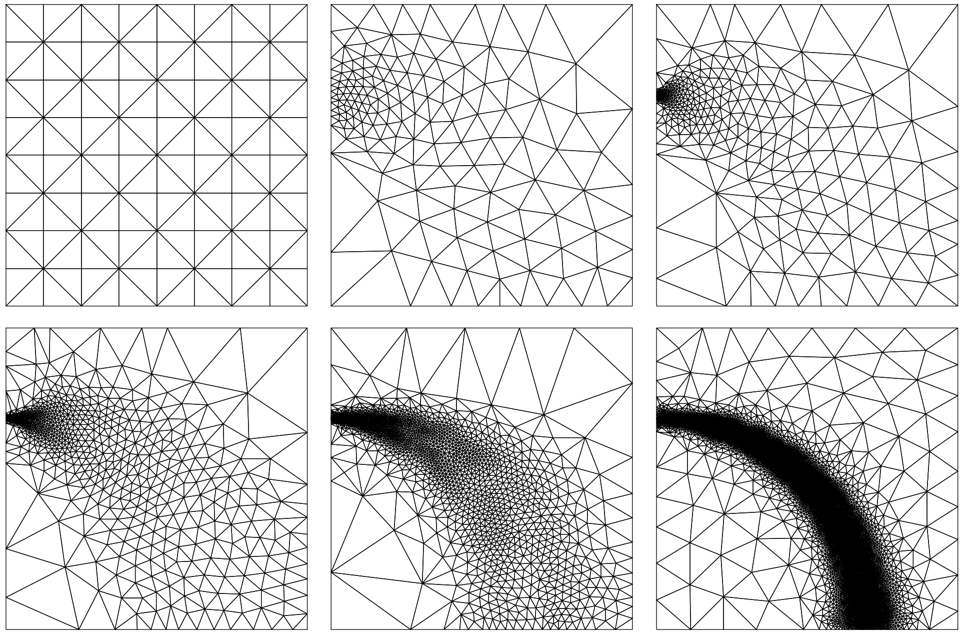

FIGURE 5. Refined meshes at some levels generated by the CfCVDT-based adaptive method for Example 2. Top-left: Initial mesh; top-middle to bottom-right: adaptively refined meshes at the refinement levels 2, 4, 6, 8, and 12.

Level ℓ	$\|e_{h,\ell}\|_{L^\infty(\Omega)}$	CR	$\|e_{h,\ell}\|_{L^2(\Omega)}$	CR	STD_η
0	6.2298e-01	–	1.6344e-01	–	1.591
1	5.8256e-01	0.54	1.3662e-01	1.43	1.340
2	5.6911e-01	0.11	1.3325e-01	0.12	1.192
3	5.6623e-01	0.02	1.1887e-01	0.54	1.330
4	5.0302e-01	0.55	1.0540e-01	0.56	1.127
5	4.9170e-01	0.09	9.2372e-02	0.53	1.181
6	4.8746e-01	0.04	7.9986e-02	0.68	1.084
7	3.8075e-01	1.04	5.6164e-02	1.61	1.014
8	3.7963e-01	0.01	3.9852e-02	1.44	1.006
9	2.9251e-01	1.09	2.5634e-02	1.82	1.023
10	2.0813e-01	1.43	1.4953e-02	2.26	1.020
11	1.5073e-01	1.36	9.2994e-03	2.00	0.991
12	9.7918e-02	1.73	5.5999e-03	2.04	0.937
13	6.4729e-02	1.58	3.3266e-03	1.99	0.948

TABLE 4. Solution errors, and convergence rates at different refinement levels for Example 2.

Upwind scheme needs to be used for discretization since this problem is convection-dominated. The initial coarse mesh used for the solution to Example 2 is a uniform Cartesian grid consisting of 81 nodes; see Fig. 5. Fig. 5 also shows adaptively refined CfCVDT meshes at some levels generated using Algorithm 1.

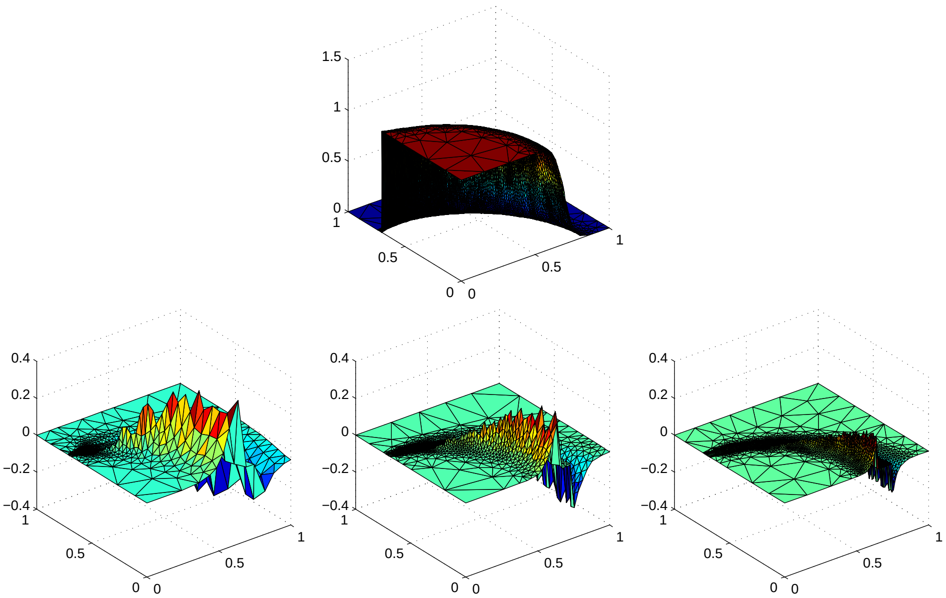

FIGURE 6. Plots of an approximate solution u_h and error distributions $e_{h,\ell}$ for Example 2. Top: plot of the approximate solution u_h at level 10; bottom (from left to right): error distributions $e_{h,\ell}$ at the refinement levels 6, 8 and 10.

The distributions of nodes in the adaptive meshes show the accumulation of nodes along the curve where the exact solution has a jump (a strong singularity). Note that while the meshes in the neighborhood of the curve are refined, the meshes apart from the curve are even coarsened when level increases since in the region away from the jump the exact solution is very smooth. Tables 3 and 4 contain information about mesh quality, solution errors, and convergence rates at different refinement levels for Example 2. The errors are measured only in the region $\{(x,y) \mid x \geq 0,\ y \geq 0,\ |\sqrt{x^2+y^2}-0.7| > 0.05|\}$ as suggested in [18] and are only under L^2 and L^∞ norms. We obtain the perfect convergence rate (order 2) in L^2 norm and almost order 1.5 of convergent rate in L^∞ norm.

The mesh size varies greatly over Ω and the variations are increasing rapidly with the level of refinement; e.g., h_{max}/h_{min} in Table 3 reaches 2447.40 at the last level; However, the values of q_{min} and q_{avg} given in Table 3 demonstrate that the shape quality of the meshes resulting from the CfCVDT-based adaptive strategy is always very good at all refinement levels. Our adaptive method still possesses the property of the equidistribution of the errors over Ω. To verify this, we plot, in Fig. 6, a representative approximate solution u_h and the errors $e_{h,\ell}$ at different refinement levels. The equidistribution of the errors over Ω can be visually verified by Fig. 6, but it does not perform as good as the first example based on the values of STD_η in Table 4. This is partially due to the fact that the exact solution has a big jump from 0 to 1 around the quarter of the circle $\{(x,y) \mid x \geq 0, y \geq 0, x^2+y^2 = 0.7^2\}$.

4.3. **Nonconvex domain with curved boundary.** The third illustrative example is given as follows.

Example 3. *Let Ω be a circular ring such as*

$$\Omega = \{(r,\theta) \mid r_1 \le r \le r_2, \quad 0 \le \theta < 2\pi\}$$

Let e_ρ and e_θ be the usual unit basis vectors of a polar coordinate system. We consider the following problem on Ω

$$-\epsilon\Delta u + e_\rho \cdot \nabla u = 0 \tag{44}$$

with boundary conditions

$$\begin{cases} u &= \dfrac{1}{r_2}\sin(\theta), \qquad \text{on} \quad \Gamma_D^2 = \{(r_2,\theta) \mid 0 \le \theta < 2\pi\}, \\ u &= \dfrac{e^{(r_1-r_2)/\epsilon}}{r_1}\sin(\theta), \qquad \text{on} \quad \Gamma_D^1 = \{(r_1,\theta) \mid 0 \le \theta < 2\pi\}. \end{cases}$$

The exact solution is

$$u = \frac{e^{(r-r_2)/\epsilon}}{r}\sin(\theta).$$

In this example, we set $\epsilon = 5.0 \times 1.0^{-3}$, $r_1 = 0.25$ and $r_2 = 1$.

The exact solution for this example is smooth but has very large gradients around the exterior boundary $\{(x,y) \mid x^2+y^2 = 1\}$ where a lot of refinements will take place [17].

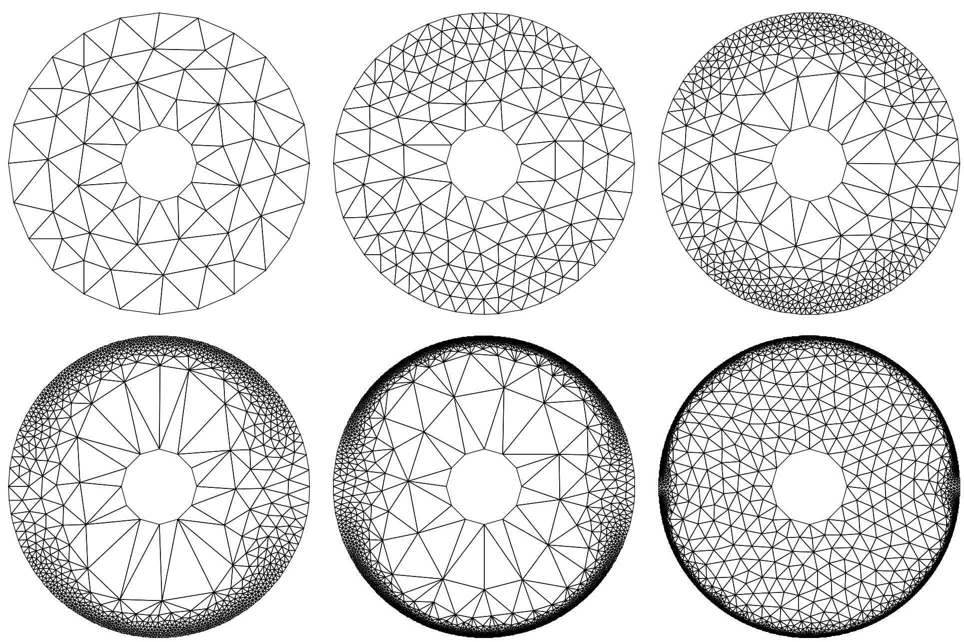

FIGURE 7. Refined meshes at some levels generated by the CfCVDT-based adaptive method for Example 3. Top-left: Initial mesh; top-left to bottom-right: adaptively refined meshes at the refinement levels 2, 4, 6, 8, and 12.

Upwind scheme again needs to be used for discretization since it is relatively convection-dominated. The initial coarse mesh with 72 nodes used for the solution of Example 3 is displayed in Fig. 7 where adaptively refined meshes at some levels generated using Algorithm 1 are also shown. Tables 5 and 6 contain the information

Level ℓ	n_ℓ	q_{min}	q_{avg}	h_{max}/h_{min}
0	72	0.639	0.889	1.85
1	107	0.572	0.917	2.44
2	172	0.619	0.923	4.02
3	276	0.582	0.928	6.41
4	444	0.541	0.929	12.53
5	704	0.437	0.930	23.41
6	1140	0.447	0.921	36.87
7	1853	0.416	0.926	56.15
8	3042	0.336	0.924	82.23
9	5072	0.342	0.925	129.43
10	8593	0.349	0.928	146.26
11	14857	0.382	0.927	173.05
12	26048	0.407	0.931	196.28

TABLE 5. Mesh quality at different refinement levels for Example 3.

Level ℓ	$\|e_{h,\ell}\|_{L^\infty(\Omega)}$	CR	$\|e_{h,\ell}\|_{L^2(\Omega)}$	CR	$\|e_{h,\ell}\|_{H^1(\Omega)}$	CR	STD_η
0	9.2372e-01	–	4.6816e-01	–	3.3128e+00	–	1.384
1	8.5826e-01	0.37	3.7617e-01	1.10	6.9024e+00	-3.71	1.263
2	8.1083e-01	0.24	2.9861e-01	0.97	1.0345e+01	-1.70	1.417
3	7.3114e-01	0.44	2.3675e-01	0.98	1.3345e+01	-1.08	1.363
4	6.5926e-01	0.44	1.7453e-01	1.28	1.4034e+01	-0.21	1.384
5	5.6539e-01	0.67	1.3009e-01	1.28	1.3617e+01	0.13	1.285
6	4.8014e-01	0.68	9.3901e-02	1.35	1.2208e+01	0.45	1.189
7	3.7852e-01	0.98	6.6251e-02	1.43	1.0302e+01	0.70	1.069
8	2.5701e-01	1.43	4.5695e-02	1.50	8.1947e+00	0.92	0.927
9	1.8128e-01	1.37	3.1273e-02	1.48	6.2242e+00	1.08	0.780
10	1.2149e-01	1.22	2.1791e-02	1.37	4.6177e+00	1.13	0.651
11	8.1699e-02	1.21	1.6621e-02	1.22	3.4000e+00	1.12	0.544
12	5.5842e-02	1.18	1.2186e-02	1.19	2.4516e+00	1.14	0.446

TABLE 6. Solution errors, and convergence rates at different refinement levels for Example 3.

about mesh quality, solution errors, and convergence rates at different levels for Example 3. It should be pointed out that we get almost similar convergence rates (about 1.2) in both L^2 and L^∞ norms and a little bit lower one in H^1 norm. The shape quality of the meshes indicated by the values of q_{min} and q_{avg} given in Table 5 are again always very good at all levels, although the mesh sizes vary greatly over Ω. The equidistribution of the errors over Ω can again be visually verified by Fig. 8; it can also be demonstrated by values of STD_η in Table 6.

5. **Discussions and conclusions.** In this paper, we presented an efficient adaptive meshing algorithm for convection-diffusion problems that combines a posteriori error estimation with centroidal Voronoi Delaunay tessellations methodology. Various numerical experiments in the two dimensional space are carried out and show that many desirable features can be achieved, including the equidistribution of errors over elements, well-shapedness of triangles at any levels and almost best convergence rates using the linear finite volume discretizations.

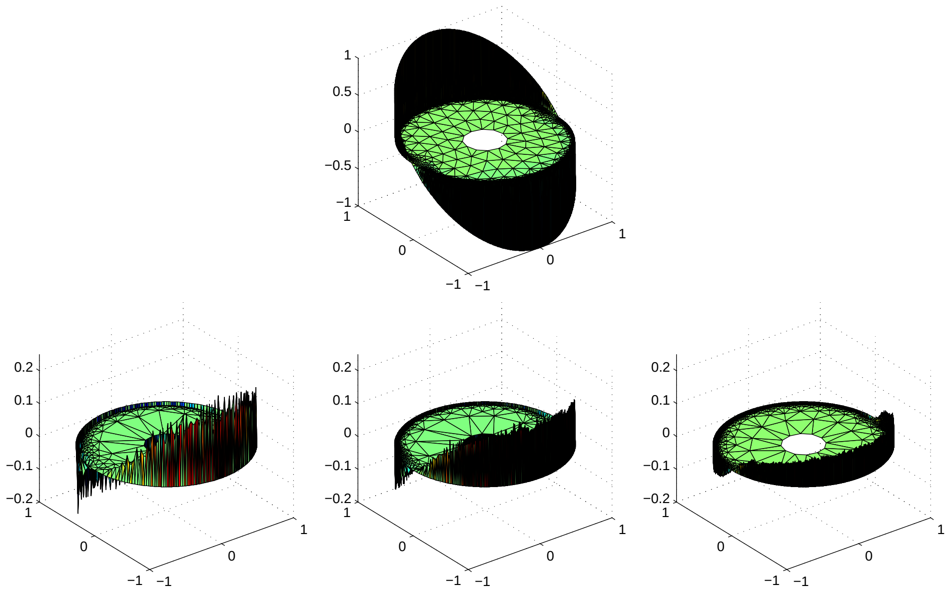

FIGURE 8. Plots of an approximate solution u_h and error distributions $e_{h,\ell}$ for Example 3. Top: plot of the approximate solution u_h at level 10; bottom (from left to right): error distributions $e_{h,\ell}$ at the refinement levels 6, 8 and 10.

Efficient implementation of three dimensional CfCVDT generation [11] is still under development. We also would like to point out that for convection-dominated problems, the most efficient meshing scheme in many cases is probably to use the anisotropic meshes instead of isotropic ones due to special properties of the solutions [30], so that adaptive meshing techniques based on the anisotropic centroidal Voronoi tessellations (ACVT) [12] is currently another major ongoing research project for us.

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